Shivam Kumar Verma

+91 91356 68614 | shivam0xverma@gmail.com | LinkedIn | GitHub | Portfolio

Experience

poiro.com **Jul 2026 – Present**
Fullstack Developer Intern *Python, FastAPI, MongoDB, Redis, Celery, Next.js, TypeScript, Cloudflare R2*

- **DAG Workflow Engine:** Built the node and group system behind a visual AI workflow builder: each workflow is a DAG of generation steps passing typed outputs between ports in execution order, and nested groups run as **sub-workflows** in one unit.
- **Digital Asset Service:** Designed the org-scoped MongoDB models and **FastAPI** backend for one asset service shared by **3 apps** (**32 routes**, keyset pagination), migrated existing assets into it, and replaced one subtree query per folder level with a single $graphLookup, cutting folder-edit database round-trips from **3 to 1** for query performance.
- **Provider Adapter Layer:** Integrated Pixabay, Unsplash, and Pinterest behind one adapter contract with per-org Redis caching and fail-closed spend caps of **500 calls/org/day** and **5,000/day** globally to stop runaway provider costs.
- **Python Codebase Health:** Cleared **1,041** Ruff findings across **7** Python packages, made lint blocking in CI, moved the baseline to Python 3.13, and refreshed **70** dependencies.

klimb.io **Nov 2025 – May 2026**
Full Stack Developer Intern *TypeScript, Express.js, PostgreSQL, Redis, AWS*

- **Usage-Based Billing Ledger:** Replaced on/off feature flags with a PostgreSQL credit ledger that meters each customer's usage of **6+ AI features**, which made usage-based pricing possible.

Projects

CodeSM | GitHub | Live | *Node.js, BullMQ, Redis, PostgreSQL, Docker*

- **Sandboxed Code Execution:** Engineered an isolated runtime for **6 languages** where each submission runs in a container with no network, a read-only root, **256 MB** memory, **32** PIDs, and a CPU and time limit, so infinite loops and fork bombs stay contained.
- **Async Grading Queue:** Moved grading off the request path onto BullMQ workers so the API returns instantly; each submission reuses **1** container across all its test cases, loaded from S3 one at a time, and exits at the first failure.

Yuvro | GitHub | Demo | *Python, FastAPI, PostgreSQL, Kubernetes, AWS S3*

- **Workspace Orchestrator:** Architected a **4**-plane cloud IDE whose **FastAPI** orchestrator provisions per-project Kubernetes deployments, services and ingress on demand, with a runner streaming PTY sessions over WebSockets.
- **Storage and Schema:** Implemented content-addressed file storage in S3 with scheduled garbage collection, over a self-referential PostgreSQL file tree with **3** composite indexes for path lookups.

DeepDoc | GitHub | *RAG, Next.js, TypeScript, Azure OpenAI, Pinecone, PostgreSQL*

- **RAG Pipeline:** Developed section-aware chunking, document-scoped vector retrieval, and follow-up rewriting from chat history, ranked and deduplicated into a token budget.
- **Evaluation Harness:** Created a **36-case** eval suite (**6** multi-turn) that sweeps **100+** retrieval configurations and separates indexing, retrieval and generation failures.

Skills

Languages: Python, TypeScript, SQL, C++
Backend: FastAPI, Celery, Node.js, Express.js, REST APIs, WebSockets, BullMQ
Data & Infrastructure: PostgreSQL, MongoDB, Redis, Pinecone, Docker, Kubernetes, AWS (EC2, S3)
AI Engineering: RAG, embeddings and vector search, LLM evaluation, AI workflow orchestration
Frontend & Tooling: React, Next.js, Git, Linux, CI pipelines

Education

Indian Institute of Information Technology Nagpur **Nov 2022 – May 2026**
B.Tech in Computer Science and Engineering *Nagpur, Maharashtra*

Achievements

- **Competitive Programming: Top 5% globally** on LeetCode (**1800+ rating**, 800+ problems solved) · **3-star** on CodeChef (1700+) · Pupil on Codeforces.
- **Hackathon: 3rd place**, Tantrafiesta 2024, IIIT Nagpur.